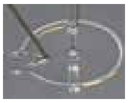

**White hole in a sink**
Scientists believe that it is possible to visualise a white hole in a sink. Read more about the experiment. **https://dgit.in/feb20-19**

**Thermodynamics Laws**
Read further about the Thermodynamics laws and the concepts they are based on. **https://dgit.in/feb20-20**

## BEYOND THE EVENT HORIZON

Since no form of light or information can escape the gravitational hold of a black hole, nobody really knows what happens inside a black hole once matter and energy fall into it. Once any matter enters the event horizon of a black hole, it would need to move at speeds faster than light to escape the gravitational pull of the black hole. Physicists believe that infalling matter undergoes a spaghettification, where they deform because of the extremes of gravitational influence. Something approximating spaghettification is seen in Interstellar when Brand interacts with the beings from the future. In fact the tidal forces would be so great while approaching a black hole, that the spaghettification can occur even beyond the event horizon. The characteristics of the region within the event horizon is said to be dependent on the exact process by which the black hole was formed. At the hearts of black holes, scientists believe that singularities could exist, points of spacetime with infinite mass and density. The laws of physics as we know them would fail to exist at the singularity. Singularities are however not a requirement for black holes to exist. Scientists have suggested all kinds of exotic hypothetical objects that can exist at the heart of black holes, including ring shaped singularities, and a spherical hypersurface – think of it as a two dimensional sheet of paper forming a three dimensional sphere. Now, the real question is where does all the matter falling into a black hole go? Truth is, nobody knows. However, one of the possible solutions is the Einstein-Rosen bridge, better known as the wormhole. Broadly speaking wormholes can be thought of as a tunnel between two points in spacetime, through which matter can exit or enter. A very specific kind of a wormhole is a tunnel between a black hole and a white hole. Matter goes in on one side, and comes out on the other. A wormhole is also theoretically possible according to Einstein's field equations. Additionally, the white hole part of the wormhole can have all kinds of exotic locations. It can be at the other side of the universe than the black hole end, it can be in another universe, and it can even be in another time. If somehow the white hole end is moved to where the black hole is, then time travel is also possible, on a purely theoretical level. No wormholes have been observed yet, and the laundry list of conditions that need to be satisfied to create a wormhole are so long and so specific, and most scientists believe the universe will never be old enough for all the factors to come together in just the right way to ever create one. Just like singularity, wormholes are also not a necessity for the existence of black holes, and the same goes for the existence of white holes as well. While the conditions that are necessary to create a black hole are relatively well understood, the exact processes that could lead to the creation of a white hole are not. One of the seemingly insurmountable difficulties involves the destruction of black holes. We just cannot do it, and cannot think of a way to do so. As white holes can be considered to be black holes in reverse time, there is no way to create a white hole, just as there is no way to destroy a black hole. However, there are a number of theories that attempt to reconcile this, and find a way in which a white hole can possibly be created. One of these suggestions involves the creation of a white hole soon after the creation of a black hole. A collapsing black hole undergoes what is known as a quantum bounce, experiencing an outward pressure from the gravitational collapse, spontaneously turning into a white hole instead of a black hole. For most of the time that black holes were studied as theoretical objects, scientists believed that nothing can escape from a black hole. However, Stephen Hawking famously predicted that black holes could emit a special type of radiation, originating from the quantum effects of the event horizon. While the Hawking radiation has not yet been proved, scientists believe that observing the particles of such a radiation could contain information on the characteristics of the interior of the black hole, the region beyond the event horizon. If at the moment of creation of a black hole, there is a finely tuned signal dispatched towards the collapsing star to interfere with the hawking radiation at the precise point of black hole formation, you would cancel it and create what is known as an eternal black hole, a black hole that does not radiate, and would stay stable for the life span of the universe. Such a black hole, in reverse time, would be a white hole. This explanation should provide some sort of an idea of exactly how precise the conditions need to be to create a white hole. If such an eternal black hole is created, it would be the perfect form of long term storage for collected wisdom or information. Hawking radiation also predicts that if mass stops falling into a black hole, then the radiation would eventually cause the black hole to shrink and disappear, albeit over extremely long periods of time. If black holes can evaporate through such a process then it implies that white holes can theoretically be created. One of the most perplexing things about white holes is that if they exist, they should be easy to observe, easier than black holes. One of the theoretical models of white holes is that at the time of creation, they just expunge all the matter and energy, and then disappear. GRB 060614 may be one such object. It was a gamma ray burst unlike any other, detected by NASA's Swift instrument in 2006. Instead of all the other gamma ray bursts that were mostly followed by the creation of a black hole, or at least a supernova remnant, GRB 060614 was just a massive a flash. Scientists still have no clue what exactly exploded to create that particular gamma ray burst. There are two types of gamma ray bursts, short and long. The short ones are mergers of two massive objects such as neutron stars or black holes. The long ones are supernovae. GRB 060614 was a long one, lasting 012 seconds, but even the Hubble space telescope could not detect a supernova remnant. The burst remains a mystery. Another theoretical kind of white hole could be the Big Bang itself. d